

Modelling Large Spatial Data

An Overview and a Deep Dive into Hierarchical Nearest-Neighbor
Gaussian Process (NNGP) Models

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① An Overview

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④ NNGP

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Background

- **Data:** Geocoded real estate transactions, along with information on several other variables like floor area, no. of bedrooms, no. of bathrooms, amenities, etc.

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- **Goal:** Modelling the transactions on the remaining variables (covariates).
- **Note:** Transactions are spatially dependent.

Spatial Model with Full Gaussian Process

$$Y(\mathbf{s}_i) = \boldsymbol{\beta}^\top \mathbf{x}(\mathbf{s}_i) + w(\mathbf{s}_i) + \varepsilon(\mathbf{s}_i), \quad i = 1, 2, \dots, N,$$

where:

- N : No. of observations
- $\mathbf{s}_i$: Geographical coordinates for i -th observation
- $Y(\mathbf{s}_i)$ is the response corresponding to the i -th observation
- $\mathbf{x}(\mathbf{s}_i)^{p \times 1}$ is the vector of covariates for the i -th observation
- $\boldsymbol{\beta}^{p \times 1}$ is the vector of fixed effects regression coefficients
- $w(\mathbf{s}_i)$ is the spatial random effect for the i -th observation
- $\varepsilon(\mathbf{s}_i)$ is random noise for the i -th observation
- $w(\mathbf{s}_i)$ and $\varepsilon(\mathbf{s}_i)$ are uncorrelated.

Spatial Model with Full Gaussian Process

- In matrix notation,

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{w} + \boldsymbol{\varepsilon},$$

where $\mathbf{Y} = (Y(\mathbf{s}_1), Y(\mathbf{s}_2), \dots, Y(\mathbf{s}_N))^T$, $\mathbf{X} = (\mathbf{x}(\mathbf{s}_1)^T, \mathbf{x}(\mathbf{s}_2)^T, \dots, \mathbf{x}(\mathbf{s}_N)^T)^T$, $\mathbf{w} = (w(\mathbf{s}_1), w(\mathbf{s}_2), \dots, w(\mathbf{s}_N))^T$ and $\boldsymbol{\varepsilon} = (\varepsilon(\mathbf{s}_1), \varepsilon(\mathbf{s}_2), \dots, \varepsilon(\mathbf{s}_N))^T$.

- $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Omega})$ and $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \tau^2 \mathbf{I}_N)$ are uncorrelated.
- Likelihood:** $\mathbf{Y} \sim \mathcal{N}(\mathbf{X}\boldsymbol{\beta}, \boldsymbol{\Sigma})$, where $\boldsymbol{\Sigma} = \boldsymbol{\Omega} + \tau^2 \mathbf{I}_N$
- Notes:**
 - Likelihood computation involves $|\boldsymbol{\Sigma}|$ and $\boldsymbol{\Sigma}^{-1}$

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- Likelihood:** $\mathbf{Y} \sim \mathcal{N}(\mathbf{X}\beta, \Sigma)$, where $\Sigma = \Omega + \tau^2 \mathbf{I}_N$
- Notes:**
 - Likelihood computation involves $|\Sigma|$ and Σ^{-1}
 - Computation time $\approx O(N^3)$, Storage $\approx O(N^2)$
 - Computationally intensive and memory prohibitive for large N

Few Remedies

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② Spatial Partitioning

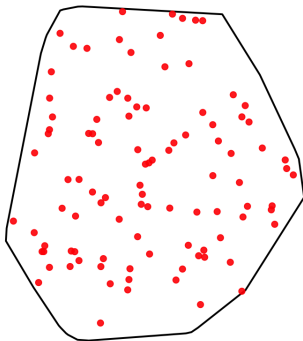
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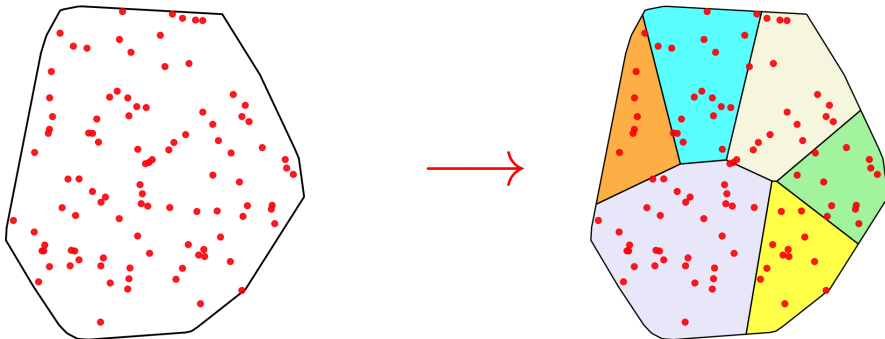
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Spatial Partitioning



Spatial Partitioning



Dividing entire spatial domain into **disjoint** and **spatially independent** regions/clusters

Spatial Partitioning

- Spatial model with full Gaussian process and likelihood:

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{w} + \boldsymbol{\varepsilon}, \quad \mathbf{Y} \sim \mathcal{N}(\mathbf{X}\boldsymbol{\beta}, \boldsymbol{\Sigma})$$

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- Partitioning of spatial domain $\mathcal{D}$ into K disjoint regions:

$$\mathcal{D} = \mathcal{D}_1 \cup \mathcal{D}_2 \cup \dots \cup \mathcal{D}_K \quad (\mathcal{D}_d \cap \mathcal{D}_{d'} = \emptyset, d \neq d')$$

Spatial Partitioning

Accordingly,

$$\mathbf{Y} = \begin{pmatrix} \mathbf{Y}_1 \\ \mathbf{Y}_2 \\ \vdots \\ \mathbf{Y}_K \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \\ \vdots \\ \mathbf{X}_K \end{pmatrix}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_K \end{pmatrix}, \quad \mathbf{w} = \begin{pmatrix} \mathbf{w}_1 \\ \mathbf{w}_2 \\ \vdots \\ \mathbf{w}_K \end{pmatrix}, \quad \boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_d \end{pmatrix}$$

- Region wise model and likelihood:

$$\mathbf{Y}_d = \mathbf{X}_d \beta_d + \mathbf{w}_d + \varepsilon_d, \quad \mathbf{Y}_d \sim \mathcal{N}(\mathbf{X}_d \beta_d, \boldsymbol{\Sigma}_d) \quad (d = 1, 2, \dots, K)$$

Spatial Partitioning

- No spatial dependence among regions:

$$\Sigma \approx \Sigma^* \begin{pmatrix} \Sigma_1 & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \Sigma_2 & \dots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \Sigma_K \end{pmatrix}$$

- Joint likelihood:

$$p(\mathbf{Y}) \approx \prod_{d=1}^K p(\mathbf{Y}_d) \quad \Rightarrow \quad \log p(\mathbf{Y}) \approx \sum_{d=1}^K \log p(\mathbf{Y}_d)$$

Spatial Partitioning

- Induces sparsity in covariance matrix through spatial independence assumption among regions.
- Region wise log-likelihoods involving inversions of region wise covariance matrices can be [computed in parallel](#).

Spatial Partitioning

- Induces sparsity in covariance matrix through spatial independence assumption among regions.
- Region wise log-likelihoods involving inversions of region wise covariance matrices can be [computed in parallel](#).
- Computation time $\approx O(N^3/K^2)$, Storage $\approx O(N^2/K)$ (for balanced partitions)

Spatial Partitioning

- **Some choices of partitions:**
 - Based on predefined regions (for example, macro markets for real estate problems)
 - Based on equal areas
 - Based on hierarchical clustering algorithms
 - Based on mixture models

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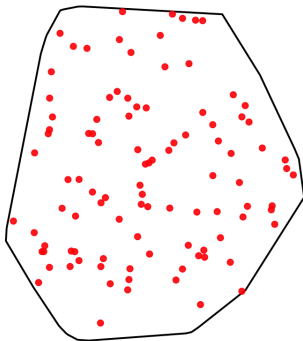
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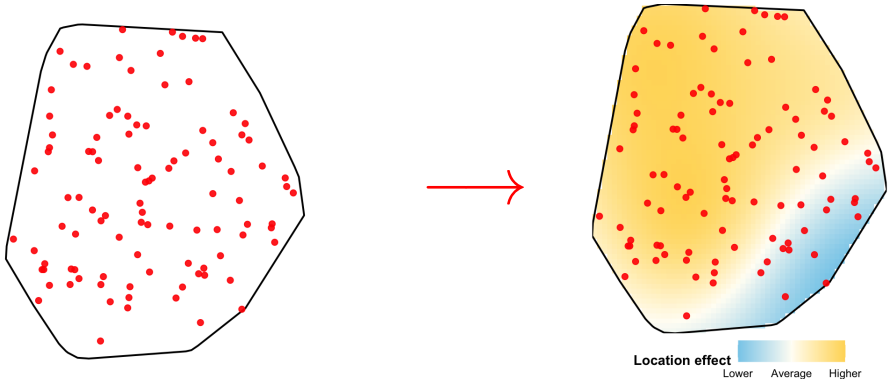
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Fixed Rank Kriging



Fixed Rank Kriging



Covering the map with **fewer overlapping smooth location effect patterns**

Fixed Rank Kriging

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$$Y(\mathbf{s}_i) = \boldsymbol{\beta}^\top \mathbf{x}(\mathbf{s}_i) + w(\mathbf{s}_i) + \varepsilon(\mathbf{s}_i), \quad i = 1, 2, \dots, N$$

Fixed Rank Kriging

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- Approximation of spatial process using basis functions:

$$w(\mathbf{s}_i) \approx \sum_{m=1}^M \eta_m h_m(\mathbf{s}_i), \quad (M \ll N)$$

where $h_1(\mathbf{s}_i), h_2(\mathbf{s}_i), \dots, h_M(\mathbf{s}_i)$ are the [basis functions](#), and $\eta_1, \eta_2, \dots, \eta_M$ are the corresponding [random basis coefficients](#).

Fixed Rank Kriging

- Model after approximation:

$$Y(\mathbf{s}_i) \approx Y^*(\mathbf{s}_i) = \boldsymbol{\beta}^\top \mathbf{x}(\mathbf{s}_i) + \sum_{m=1}^M \eta_m h_m(\mathbf{s}_i) + \varepsilon(\mathbf{s}_i), \quad i = 1, 2, \dots, N$$

Fixed Rank Kriging

- Joint model of all observations:

$$\mathbf{Y} \approx \mathbf{Y}^* = \mathbf{X}\boldsymbol{\beta} + \mathbf{H}\boldsymbol{\eta} + \boldsymbol{\varepsilon},$$

$$\text{where } \mathbf{Y}^* = \begin{pmatrix} Y^*(\mathbf{s}_1) \\ Y^*(\mathbf{s}_2) \\ \vdots \\ Y^*(\mathbf{s}_N) \end{pmatrix}, \boldsymbol{\eta} = \begin{pmatrix} \eta_1 \\ \eta_2 \\ \vdots \\ \eta_M \end{pmatrix}, \mathbf{H} = \begin{pmatrix} h_1(\mathbf{s}_1) & h_2(\mathbf{s}_1) & \dots & h_M(\mathbf{s}_1) \\ h_1(\mathbf{s}_2) & h_2(\mathbf{s}_2) & \dots & h_M(\mathbf{s}_2) \\ \vdots & \vdots & \vdots & \vdots \\ h_1(\mathbf{s}_N) & h_2(\mathbf{s}_N) & \dots & h_M(\mathbf{s}_N) \end{pmatrix}.$$

Fixed Rank Kriging

- **Covariance Structure:**

- $\Sigma \approx \Sigma^* = \text{Var}(\mathbf{Y}^*) = \mathbf{H}\mathbf{\Lambda}\mathbf{H}^\top + \tau^2\mathbf{I}_N$, where $\mathbf{\Lambda}^{M \times M} = \text{Var}(\boldsymbol{\eta})$ has rank $M \ll N$

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- $\Sigma^{*-1} = (\mathbf{H}\mathbf{\Lambda}\mathbf{H}^\top + \tau^2\mathbf{I}_N)^{-1} = (\tau^2)^{-1} [\mathbf{I}_N - \mathbf{H}(\tau^2\mathbf{\Lambda}^{-1} + \mathbf{H}^\top\mathbf{H})^{-1}\mathbf{H}^\top]$
- $|\Sigma^*| = (\tau^2)^N |\mathbf{I}_M + (\tau^2)^{-1}\mathbf{\Lambda}\mathbf{H}^\top\mathbf{H}|$

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- $|\Sigma^*| = (\tau^2)^N |\mathbf{I}_M + (\tau^2)^{-1}\mathbf{\Lambda}\mathbf{H}^\top\mathbf{H}|$

Likelihood computation only involves inversions & determinants of $M \times M$ matrices

Fixed Rank Kriging

- Induces **low rank approximation** in the covariance matrix of spatial process by representing the spatial process through **linear combination of basis functions**.

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- Computation time $\approx O(NM^2)$, Storage $\approx O(NM)$

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Recall

Full GP spatial regression model

For observations collected at locations $s_1, \dots, s_n$,

$$y(s_i) = \mathbf{x}(s_i)^\top \boldsymbol{\beta} + w(s_i) + \varepsilon(s_i), \quad \mathbf{w} \sim N(\mathbf{0}, \mathbf{C}(\boldsymbol{\theta})).$$

Covariance structure

$$\mathbf{C}(\boldsymbol{\theta}) = \begin{bmatrix} C_{11} & \cdots & C_{1n} \\ \vdots & \ddots & \vdots \\ C_{n1} & \cdots & C_{nn} \end{bmatrix}_{n \times n}$$

$$C_{ij} = C(s_i, s_j \mid \boldsymbol{\theta}) = \text{Cov}\{w(s_i), w(s_j)\}$$

Computational cost

Model estimation involves

$$\mathbf{C}(\boldsymbol{\theta})^{-1} \quad \text{and} \quad |\mathbf{C}(\boldsymbol{\theta})|.$$

$$\text{Flops} \approx \mathcal{O}(n^3), \quad \text{Storage} \approx \mathcal{O}(n^2).$$

Becomes infeasible for large n .

Other Approaches for Large Spatial Data

Low-rank approximation

Main idea

Represent the spatial process at all n locations using its values at a much smaller set of $r \ll n$ representative locations, called **knots**.

Computational cost : $\mathcal{O}(nr^2)$.

Advantage

Fast when the number of knots r is small.

Limitation

With too few knots, the fitted spatial surface may be overly smooth and may miss important local variation. Increasing r improves the approximation but also increases the computational cost.

Sparsity-inducing methods

Main idea

Retain the large spatial process, but make many entries of the covariance matrix or precision matrix equal to zero. Thus, dependence is represented mainly through nearby locations.

Advantage

Sparse matrices require much less storage and allow faster computations while preserving local spatial structure.

Limitation

Some earlier sparse methods provide only a finite-dimensional likelihood approximation. They may not define a complete spatial process for inference and prediction at arbitrary new locations.

What Gap Does NNGP Fill?

Problem with existing approaches

- Low-rank methods may lose fine-scale spatial variation.
- Earlier sparse methods often used sparsity for parameter estimation, while prediction at new locations required a separate kriging step.

NNGP contribution

NNGP defines a **valid spatial process** while retaining sparsity.

Hence, estimation, spatial process inference, and prediction are all performed within **one unified model**.

Basic Notation

Consider a q -variate spatial process over $\mathbb{R}^d$:

$$\mathbf{w}(\mathbf{s}) \sim GP(\mathbf{0}, \mathbf{C}(\cdot, \cdot \mid \boldsymbol{\theta})), \quad \mathbf{s} \in \mathcal{D} \subseteq \mathbb{R}^d.$$

- $\mathcal{D}$: spatial domain of interest.
- d : dimension of the spatial domain (usually $d = 2$ for longitude–latitude).
- $\mathbf{s}$: a location in the spatial domain.
- q : number of spatial variables/processes.
-

$$\mathbf{w}(\mathbf{s}) = (w_1(\mathbf{s}), \dots, w_q(\mathbf{s}))^\top \in \mathbb{R}^q$$

is the q -dimensional spatial process at location $\mathbf{s}$.

- $\boldsymbol{\theta}$: parameters associated with the spatial covariance structure.

Reference Set Notation

Let

$$\mathcal{S} = \{\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_k\}$$

be a fixed set of k distinct locations, called the **reference set**.

Define

$$\mathbf{w}_{\mathcal{S}} = \left(\mathbf{w}(\mathbf{s}_1)^{\top}, \dots, \mathbf{w}(\mathbf{s}_k)^{\top} \right)^{\top},$$

so that $\mathbf{w}_{\mathcal{S}}$ is $qk \times 1$.

$$\mathbf{w}_{\mathcal{S}} \sim N(\mathbf{0}, \mathbf{C}_{\mathcal{S}}(\theta)),$$

where $\mathbf{C}_{\mathcal{S}}(\theta)$ is a $qk \times qk$ positive-definite block covariance matrix with $\mathbf{C}(\mathbf{s}_i, \mathbf{s}_j)$ as its blocks.

NNGP Idea: Conditional Density Approximation

The full joint density can be written as a product of conditional densities:

$$p(\mathbf{w}_S) = p(\mathbf{w}(\mathbf{s}_1))p(\mathbf{w}(\mathbf{s}_2) \mid \mathbf{w}(\mathbf{s}_1)) \cdots p(\mathbf{w}(\mathbf{s}_k) \mid \mathbf{w}(\mathbf{s}_{k-1}), \dots, \mathbf{w}(\mathbf{s}_1)).$$

Problem: for large k , the conditioning set becomes very large.

In NNGP approach: replace the large conditioning set by a small neighbor set $N(\mathbf{s}_i)$, with size at most m , where $m \ll k$.

$$\tilde{p}(\mathbf{w}_S) = \prod_{i=1}^k p(\mathbf{w}(\mathbf{s}_i) \mid \mathbf{w}_{N(\mathbf{s}_i)}).$$

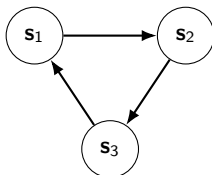
Here, $\mathbf{w}_{N(\mathbf{s}_i)}$ is the vector of spatial process values at the selected neighbors of $\mathbf{s}_i$.

Key idea: each location depends only on a few nearest previous locations.

Neighbor Sets as a Directed Graph

Idea: Treat locations as nodes. If $\mathbf{s}_j \in N(\mathbf{s}_i)$, draw a directed edge from $\mathbf{s}_i$ to $\mathbf{s}_j$.

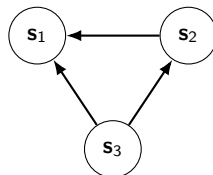
Directed cyclic graph



There is a closed directed path:

$$\mathbf{s}_1 \rightarrow \mathbf{s}_2 \rightarrow \mathbf{s}_3 \rightarrow \mathbf{s}_1.$$

Directed acyclic graph



No closed directed path. This is a directed acyclic graph.

In NNGP, neighbor sets are chosen so that the graph is acyclic. This ensures that the product of conditional densities defines a valid joint density.

When $p(\mathbf{w}_S)$ is multivariate Gaussian density then,

$$\tilde{p}(\mathbf{w}_S) = \prod_{i=1}^k N(\mathbf{w}(\mathbf{s}_i) \mid \mathbf{B}_{\mathbf{s}_i} \mathbf{w}_{N(\mathbf{s}_i)}, \mathbf{F}_{\mathbf{s}_i}) .$$

where

$$\mathbf{B}_{\mathbf{s}_i} = \mathbf{C}_{\mathbf{s}_i, N(\mathbf{s}_i)} \mathbf{C}_{N(\mathbf{s}_i)}^{-1},$$

$$\mathbf{F}_{\mathbf{s}_i} = \mathbf{C}(\mathbf{s}_i, \mathbf{s}_i) - \mathbf{C}_{\mathbf{s}_i, N(\mathbf{s}_i)} \mathbf{C}_{N(\mathbf{s}_i)}^{-1} \mathbf{C}_{N(\mathbf{s}_i), \mathbf{s}_i} .$$

- $\mathbf{B}_{\mathbf{s}_i} \mathbf{w}_{N(\mathbf{s}_i)}$: conditional mean of $\mathbf{w}(\mathbf{s}_i)$ given its neighbors.
- $\mathbf{F}_{\mathbf{s}_i}$: conditional covariance, i.e., remaining uncertainty after conditioning on the neighbors.
- $N(\mathbf{s}_i)$ contains at most m members for each $\mathbf{s}_i \in S$, where $m \ll k$, then $\tilde{\mathbf{C}}_S^{-1}$ is sparse with at most $\frac{k m(m+1)q^2}{2}$ non-zero entries.

Extending the Model Beyond the Reference Set

So far, the joint density is defined only over the reference set

$$\mathcal{S} = \{\mathbf{s}_1, \dots, \mathbf{s}_k\}.$$

Now consider a new location

$$\mathbf{u} \notin \mathcal{S}.$$

Let $N(\mathbf{u})$ be the set of the m nearest neighbors of $\mathbf{u}$ from $\mathcal{S}$. Then,

$$\mathbf{w}(\mathbf{u}) \mid \mathbf{w}_{\mathcal{S}} \sim N(\mathbf{B}_{\mathbf{u}}\mathbf{w}_{N(\mathbf{u})}, \mathbf{F}_{\mathbf{u}}).$$

where

$$\mathbf{B}_{\mathbf{u}} = \mathbf{C}_{\mathbf{u}, N(\mathbf{u})} \mathbf{C}_{N(\mathbf{u})}^{-1}, \quad \mathbf{F}_{\mathbf{u}} = \mathbf{C}(\mathbf{u}, \mathbf{u}) - \mathbf{C}_{\mathbf{u}, N(\mathbf{u})} \mathbf{C}_{N(\mathbf{u})}^{-1} \mathbf{C}_{N(\mathbf{u}), \mathbf{u}}.$$

Thus, a new location depends only on its m nearest reference locations.

Now, consider a finite set of locations outside $\mathcal{S}$,

$$\mathcal{U} = \{\mathbf{u}_1, \dots, \mathbf{u}_r\}, \quad \mathcal{U} \cap \mathcal{S} = \emptyset.$$

The conditional density is defined as

$$\tilde{p}(\mathbf{w}_{\mathcal{U}} \mid \mathbf{w}_{\mathcal{S}}) = \prod_{i=1}^r p(\mathbf{w}(\mathbf{u}_i) \mid \mathbf{w}_{N(\mathbf{u}_i)}).$$

Under the parent Gaussian process,

$$\mathbf{w}_{\mathcal{U}} \mid \mathbf{w}_{\mathcal{S}} \sim N(\mathbf{B}_{\mathcal{U}} \mathbf{w}_{\mathcal{S}}, \mathbf{F}_{\mathcal{U}}),$$

where

$$\mathbf{F}_{\mathcal{U}} = \text{diag}(\mathbf{F}_{\mathbf{u}_1}, \dots, \mathbf{F}_{\mathbf{u}_r}).$$

$\mathbf{B}_{\mathcal{U}}$ is sparse matrix of order $rq \times kq$ with each row having at most mq non-zero entries.

NNGP

For any finite set $\mathcal{V} = \{\mathbf{v}_1, \dots, \mathbf{v}_r\} \subset \mathcal{D}$, NNGP defines

$$\mathbf{w}_{\mathcal{V}} \sim N(\mathbf{0}, \tilde{\mathbf{C}}_{\mathcal{V}})$$

where $\tilde{\mathbf{C}}_{\mathcal{V}}$ is formed from pairwise covariances $\tilde{\mathbf{C}}(\mathbf{v}_i, \mathbf{v}_j)$.

$$\tilde{\mathbf{C}}(\mathbf{v}_1, \mathbf{v}_2) = \begin{cases} \tilde{\mathbf{C}}_{\mathbf{s}_i, \mathbf{s}_j}, & \mathbf{v}_1, \mathbf{v}_2 \in \mathcal{S}, \\ \mathbf{B}_{\mathbf{v}_1} \tilde{\mathbf{C}}_{N(\mathbf{v}_1), \mathbf{s}_j}, & \mathbf{v}_1 \notin \mathcal{S}, \mathbf{v}_2 = \mathbf{s}_j \in \mathcal{S}, \\ \mathbf{B}_{\mathbf{v}_1} \tilde{\mathbf{C}}_{N(\mathbf{v}_1), N(\mathbf{v}_2)} \mathbf{B}_{\mathbf{v}_2}^{\top} + \delta_{\mathbf{v}_1 = \mathbf{v}_2} \mathbf{F}_{\mathbf{v}_1}, & \mathbf{v}_1, \mathbf{v}_2 \notin \mathcal{S}. \end{cases}$$

Covariance intuition

- **Both in $\mathcal{S}$:** use the covariance already defined on the reference set.
- **One outside $\mathcal{S}$:** the outside location is connected to the reference set through its nearest neighbors.
- **Both outside $\mathcal{S}$:** their dependence is passed through the covariance between their neighbor sets.

Computational Complexity of NNGP

NNGP avoids operations on the full $n \times n$ covariance matrix.

For each location, it uses only its m nearest neighbors, where

$$m \ll n.$$

Hence, only small $m \times m$ matrix operations are required.

The dominant computational cost is

$$\mathcal{O}\left((n+k)m^3\right)$$

where k is the size of the reference set.

If m is fixed and small, the cost grows approximately linearly with the number of locations.

Choice of Reference Set $\mathcal{S}$ and Neighbor Size m

Choice of reference set $\mathcal{S}$

- A simple and effective choice is $\mathcal{S} = \mathcal{T}$, where $\mathcal{T}$ is the set of observed locations.

Choice of neighbor size m

- Values between 10 and 15 generally performed close to the full GP.

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Simulation Experiment: Setup

Data generation

- Generated data at $n = 2500$ locations in a unit square block.
- Used a univariate spatial regression model with:

$$y(s_i) = \beta_0 + \beta_1 x_i + w(s_i) + \varepsilon_i.$$

- Covariate $x_i \sim N(0, 1)$.
- Spatial effect $w(s)$ generated from a Gaussian process with exponential covariance.

Models compared

- Full Gaussian Process model.
- NNGP models with different neighbor sizes m .
- NNGP with $S = T$ and $S \neq T$.
- Gaussian Predictive Process with 64 knots (a low rank approximation method).

Simulation Result

		NNGP ($S \neq \mathcal{T}$)		NNGP ($S = \mathcal{T}$)	
		$m = 10, k = 2000$	$m = 20, k = 2000$	$m = 10$	$m = 20$
β_0	1	0.99 (0.71, 1.48)	1.02 (0.73, 1.49)	1.00 (0.62, 1.31)	1.03 (0.65, 1.34)
β_1	5	5.00 (4.98, 5.03)	5.01 (4.98, 5.03)	5.01 (4.99, 5.03)	5.01 (4.99, 5.03)
σ^2	1	1.09 (0.89, 1.49)	1.04 (0.85, 1.40)	0.96 (0.78, 1.23)	0.94 (0.77, 1.20)
τ^2	0.1	0.07 (0.04, 0.10)	0.07 (0.04, 0.10)	0.10 (0.08, 0.13)	0.10 (0.08, 0.13)
ϕ	12	11.81 (8.18, 15.02)	12.21 (8.83, 15.62)	12.93 (9.70, 16.77)	13.36 (9.99, 17.15)
p_D	—	1491.08	1478.61	1243.32	1249.57
DiC	—	1856.85	1901.57	2390.65	2377.51
G	—	33.67	35.68	77.84	76.40
P	—	253.03	259.13	340.40	337.88
D	—	286.70	294.82	418.24	414.28
RMSPE	—	1.22	1.22	1.2	1.2
95% CI cover %	—	97.2	97.2	97.6	97.6
95% CI width	—	2.19	2.18	2.13	2.12
Time	—	14.2	47.08	9.98	33.5
		Predictive process 64 knots	Full Gaussian process		
β_0	1	1.30 (0.54, 2.03)	1.03 (0.69, 1.34)		
β_1	5	5.03 (4.99, 5.06)	5.01 (4.99, 5.03)		
σ^2	1	1.29 (0.96, 2.00)	0.94 (0.76, 1.23)		
τ^2	0.1	0.08 (0.04, 0.13)	0.10 (0.08, 0.12)		
ϕ	12	5.61 (3.48, 8.09)	13.52 (9.92, 17.50)		
p_D	—	1258.27	1260.68		
DiC	—	13677.97	2364.80		
G	—	1075.63	74.80		
P	—	200.39	333.27		
D	—	1276.03	408.08		
RMSPE	—	1.68	1.2		
95% CI cover %	—	95.6	97.6		
95% CI width	—	2.97	2.12		
Time	—	43.36	560.31		

Figure 1: Univariate synthetic data analysis parameter estimates and computing time in minutes for NNGP and full GP models.

Simulation Result

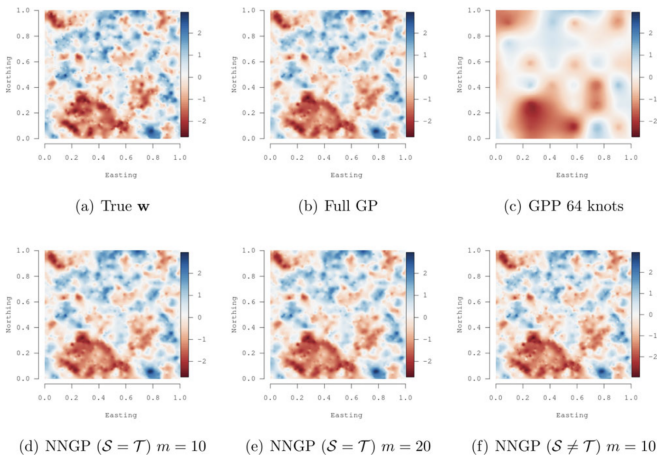


Figure 2: Univariate synthetic data analysis: interpolated surfaces of the true spatial random effects and posterior median estimates for different models.

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Thank You